

Activity 4

Aim: To obtain pure water from impure water containing ink by simple distillation method.

Apparatus: Distilling flask, water condenser, thermometer, retort stand with clamps, Bunsen burner, a receiver (flask), etc.

Chemical: Impure water containing ink.

Theory: Distillation method is generally used to purify a volatile liquid from a non volatile substance dissolved in it, or from a liquids which are less readily volatile than the liquid to be purified.

A pure liquid under constant pressure boils at a constant temperature and hence distillation of a liquid forms a method of purifying it. This method is useful however only when the liquid does not decompose on boiling.

The method consists of heating the liquid substance in a flask known as the distilling flask (fig. 4.1). The flask is heated directly on a wire gauze by a Bunsen burner or indirectly in a water/oil bath. The liquid boils and forms vapors, leaving behind the impurities in the distilling flask. The vapors of the liquid is then cooled in a water condenser, where it condenses to reproduce the liquid in a pure form. The liquid collects is a flask known as the receiver.

The distilling flask is normally a round bottom flask with a side tube (arm). A cork carrying a thermometer through it is fitted to the distilling flask so that the bulb of the thermometer is in level with the side tube. The side tube is connected to the water condenser by another cork (adapter). Cold water is circulated through the condenser in a direction opposite to the flow of the vapor of the liquid substance. This cold water keeps the condenser cool and provides a cool surface for the vapors to condense.

The temperature at which vapors start to condense in liquid and remains constant, is nothing but the boiling point of water. This temperature in the thermometer remains constant till all distillate will collect into receiver.

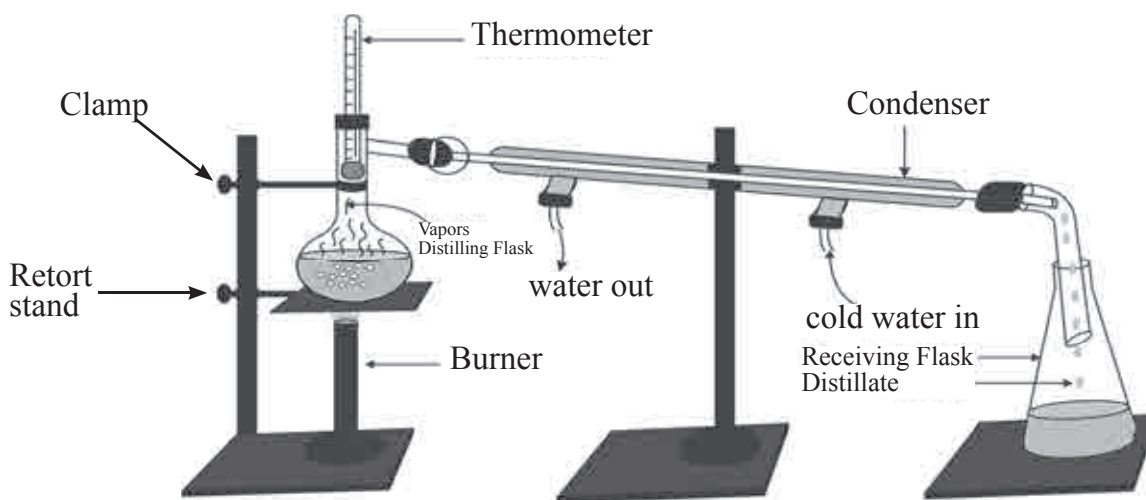


Fig. Activity 4.1 Simple distillation

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Procedure:

1. Pour impure water containing ink into the distilling flask by means of a funnel with a long stem. Seal the mouth of the distilling flask with rubber cork having a hole at the center in which thermometer is fitted. Add some pieces of porcelain to avoid bumping.
2. Now heat the flask carefully by means of a Bunsen burner. The solution begins to boil after some time.
3. The water vapors formed passes into the condenser through the side tube of the distilling flask.
4. The vapors are cooled and condensed to drops of water which collect into the receiver.
5. The pure water obtained is distilled water (distillate) having no ink contain.
6. Impurities (ink) remains behind in distilling flask.

Conclusion/ Result: The pure water is obtained from the impure water.

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. The boiling point of pure water is
a. 100 K b. 273 K ✓ c. 373 K d. 273 °C
2. Impure organic liquid having soluble impurities can be purified by a
a. crystallization ✓ c. simple distillation
b. fractional crystallization d. fractional distillation
3. ✓ Which of the following statement is correct about simple distillation
a. ✓ process to convert impure liquid into pure form.
b. process to convert pure liquid into impure liquid.
c. process to convert impure solid in pure form.
d. process to separate mixture of two different impure liquid.

Short answer questions

1. Define simple distillation method?

Ans. Simple distillation method is generally used to purify a volatile liquid from a non-volatile substance dissolved in it or from liquids which are less readily volatile than the liquid to be purified.

2. What is the role of water condenser in simple distillation method?

Ans. The condenser cools vapours of the boiling liquid, condensing them back to liquid in pure form.

3. What is the role of bumping stone in simple distillation method?

Ans. These stones have pores inside which provide cavities to trap air and spaces where bubbles of solvent vapour can form. These bubbles ensure even boiling and prevent bumping and boiling over of the solution.

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